Bayesian Basket Trial Design with False Discovery Rate Control

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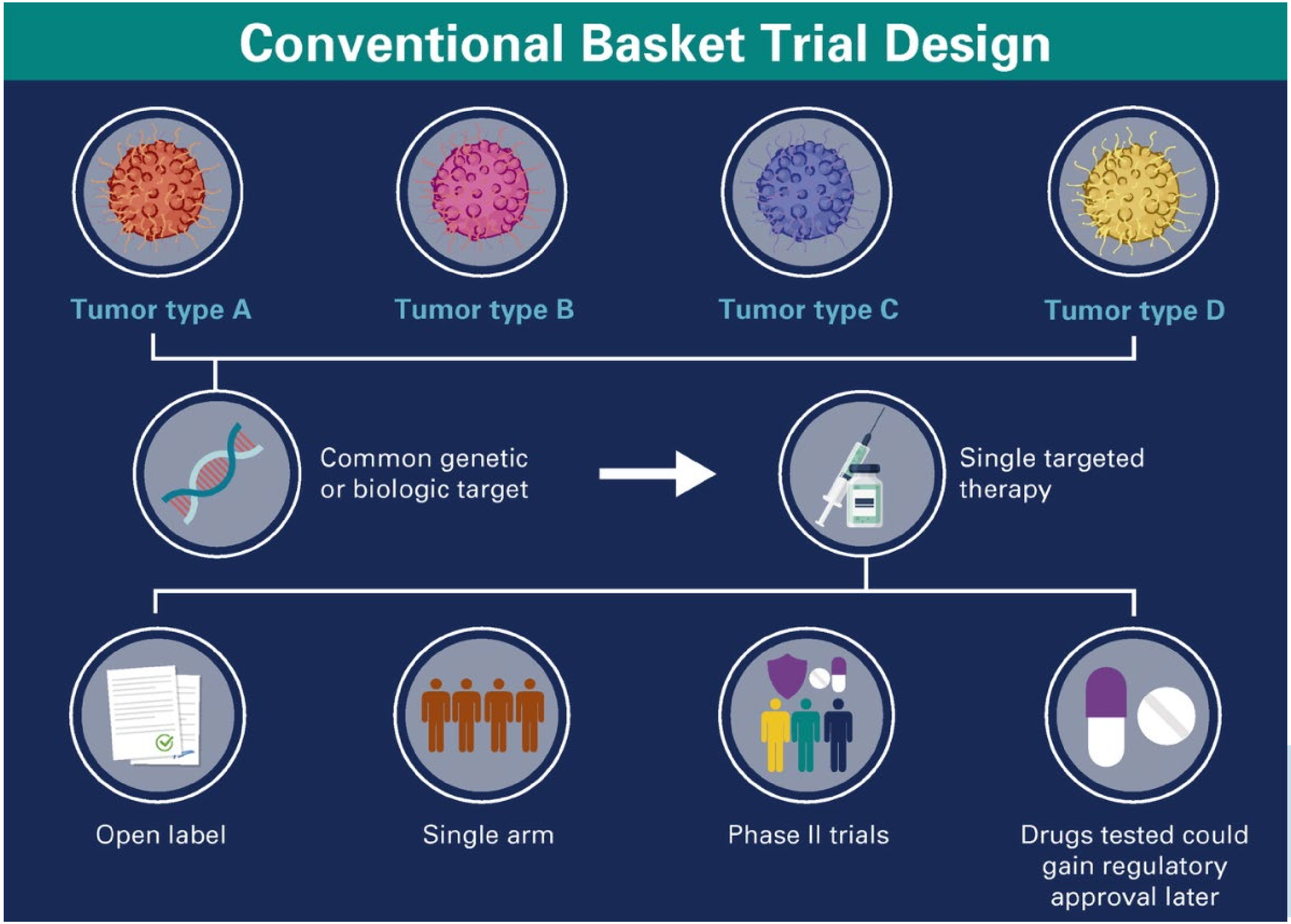
1. **Umbrella trials**: study multiple targeted therapies in the context of a single disease
2. **Basket trials**: study a single targeted therapy in the context of multiple diseases or disease subtypes
3. **Platform trials**: study multiple targeted therapies in the context of a single disease in an adaptive, perpetual manner

Woodcock, J., & LaVange, L. M. (2017). Master Protocols to Study Multiple Therapies, Multiple Diseases, or Both. *The New England journal of medicine*, *377*(1), 62–70.



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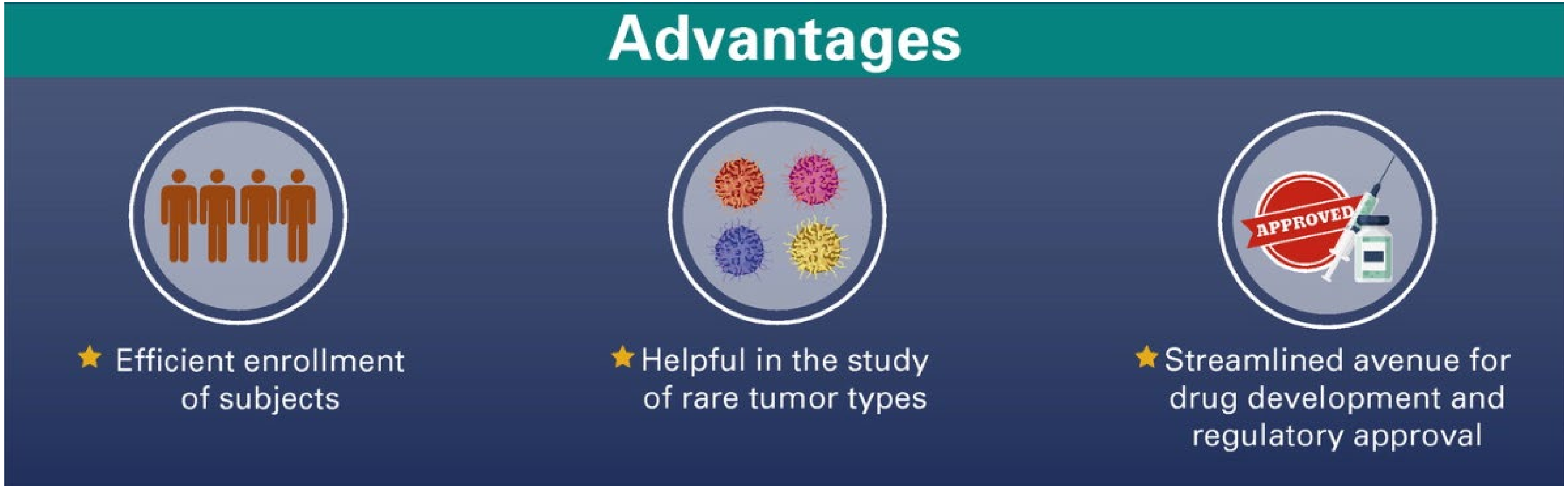

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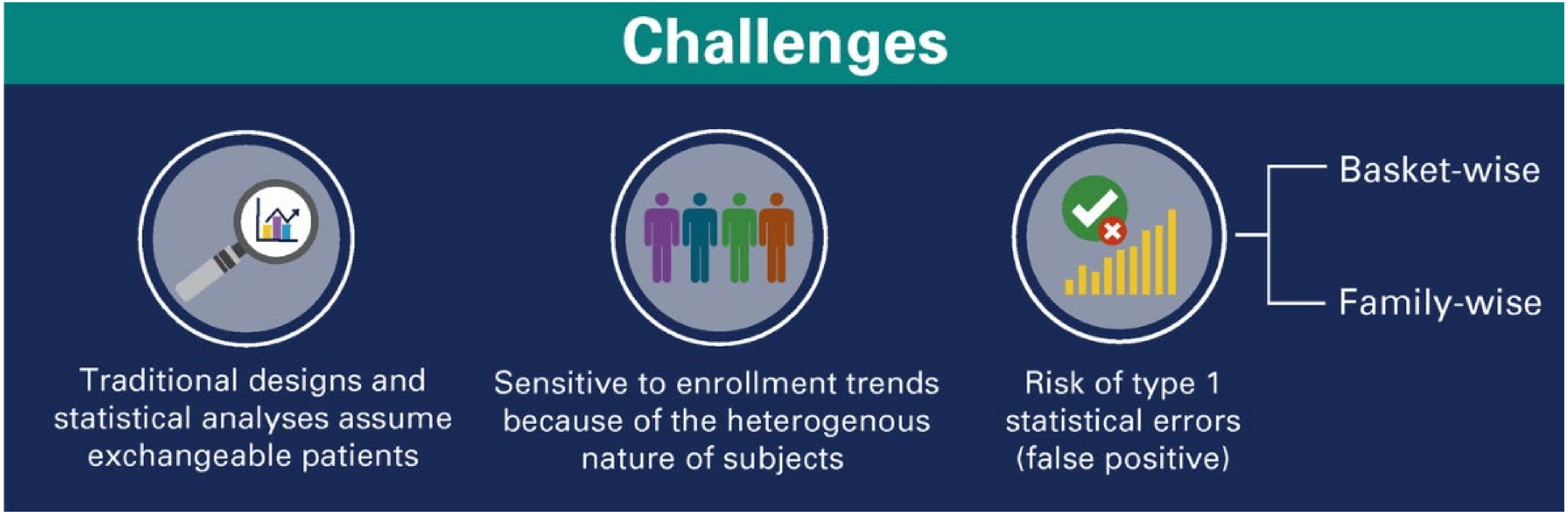

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Pooled analysis across all baskets may miss heterogeneity signals, and ignores subtype-specific differences in outcomes.

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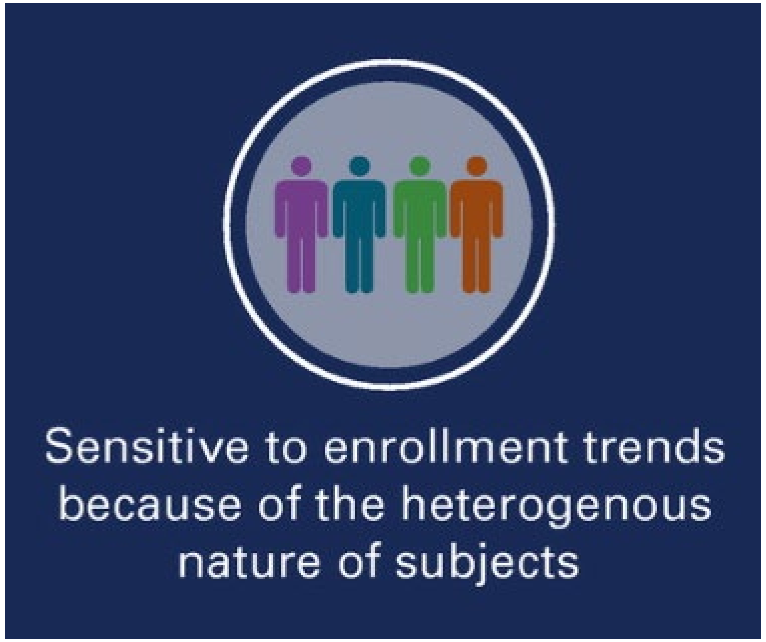

Enrollment trends may preclude evaluation of baskets with low enrollment or lead to erroneous conclusions for sparsely enrolled tumor types in the presence of heterogeneity.

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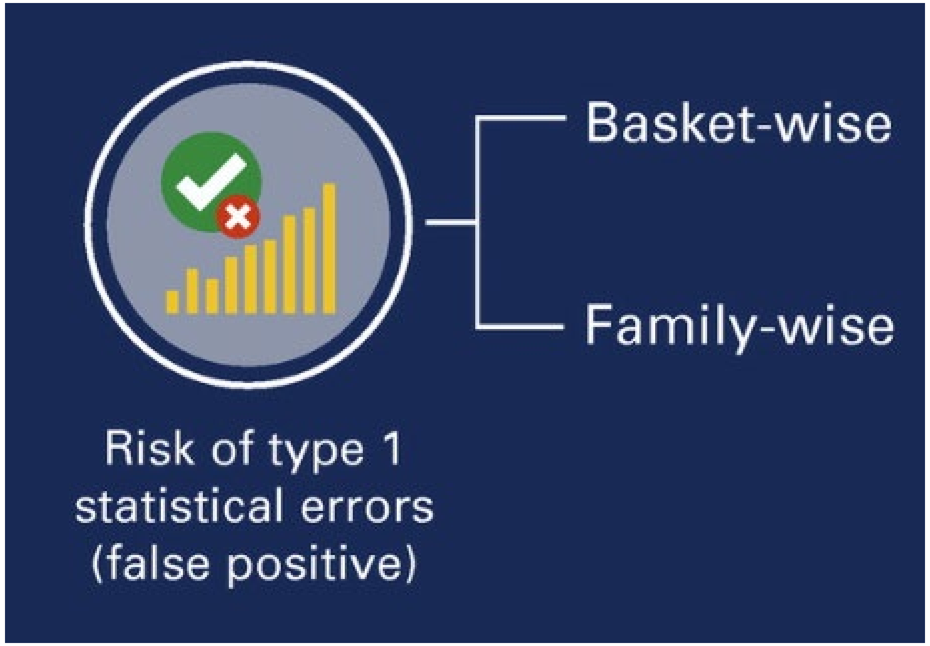

Independent analysis of each basket **inflates the type I error rate**.

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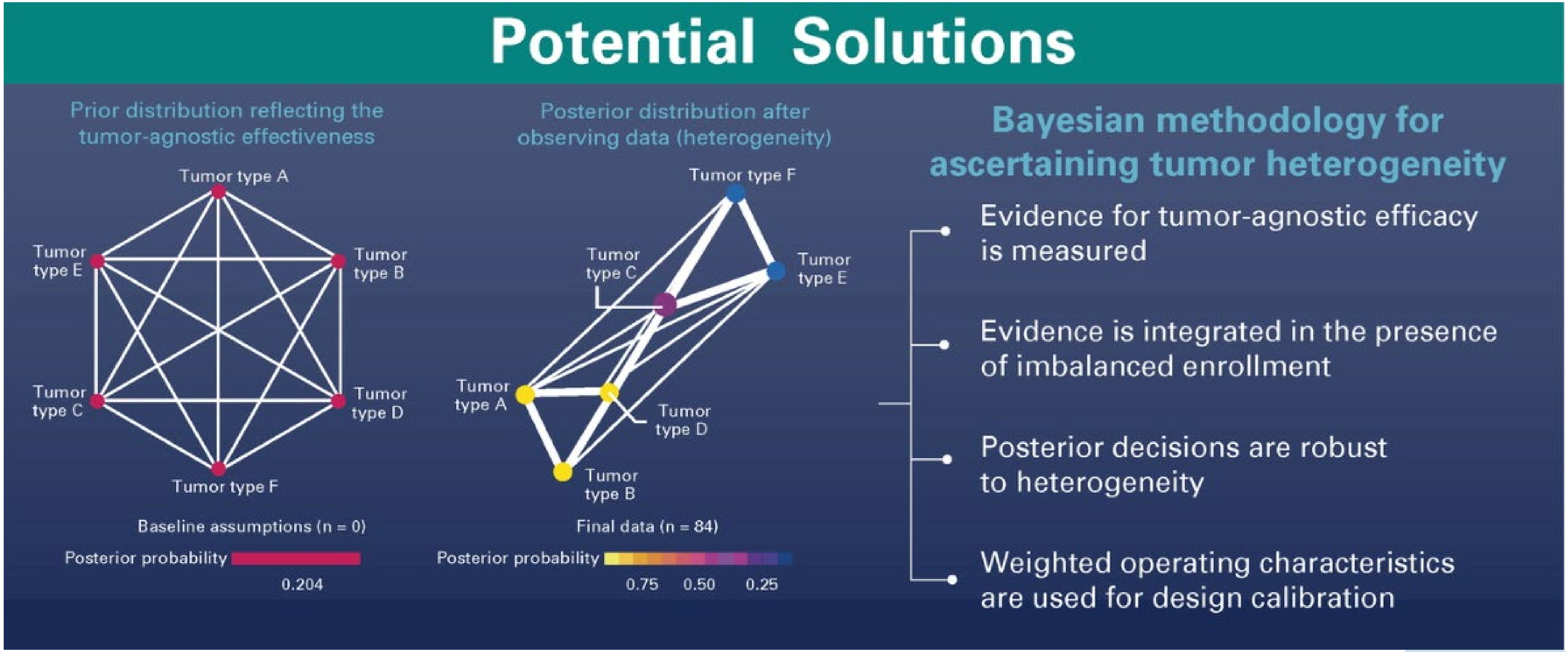

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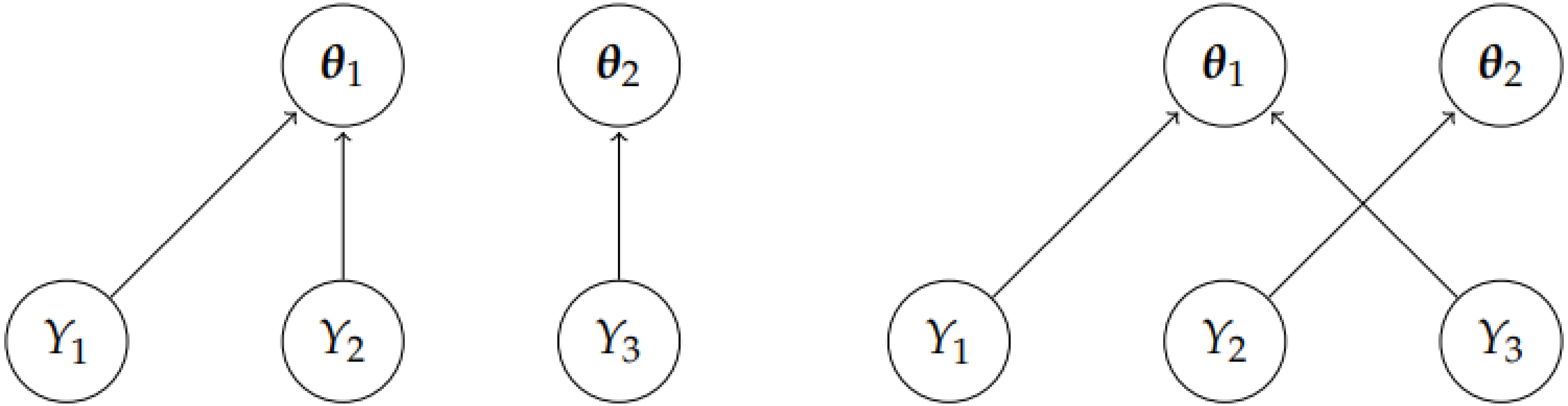
The multisource exchangeability model (MEM):



(a) Model where Y_1 and Y_2 are exchangeable. **(b)** Model where Y_1 and Y_3 are exchangeable.

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Hobbs, B. P., & Landin, R. (2018). Bayesian basket trial design with exchangeability monitoring. *Statistics in medicine*, *37*(25), 3557–3572.

basket

lifecycle stable | build passing | build failing | codecov 78% | R-CMD-check failing

Basket designs are prospective clinical trials that are devised with the hypothesis that the presence of selected molecular features determine a patient's subsequent response to a particular "targeted" treatment strategy. Basket trials are designed to enroll multiple clinical subpopulations to which it is assumed that the therapy in question offers beneficial efficacy in the presence of the targeted molecular profile. The treatment, however, may not offer acceptable efficacy to all subpopulations enrolled. Moreover, for rare disease settings, such as oncology wherein these trials have become popular, marginal measures of statistical evidence are difficult to interpret for sparsely enrolled subpopulations. Consequently, basket trials pose challenges to the traditional paradigm for trial design, which assumes inter-patient exchangeability. The R-package `basket` facilitates the analysis of basket trials by implementing multi-source exchangeability models. By evaluating all possible pairwise exchangeability relationships, this hierarchical modeling framework facilitates Bayesian posterior shrinkage among a collection of discrete and pre-specified subpopulations.

Installation

You can install the released version of basket from CRAN with:

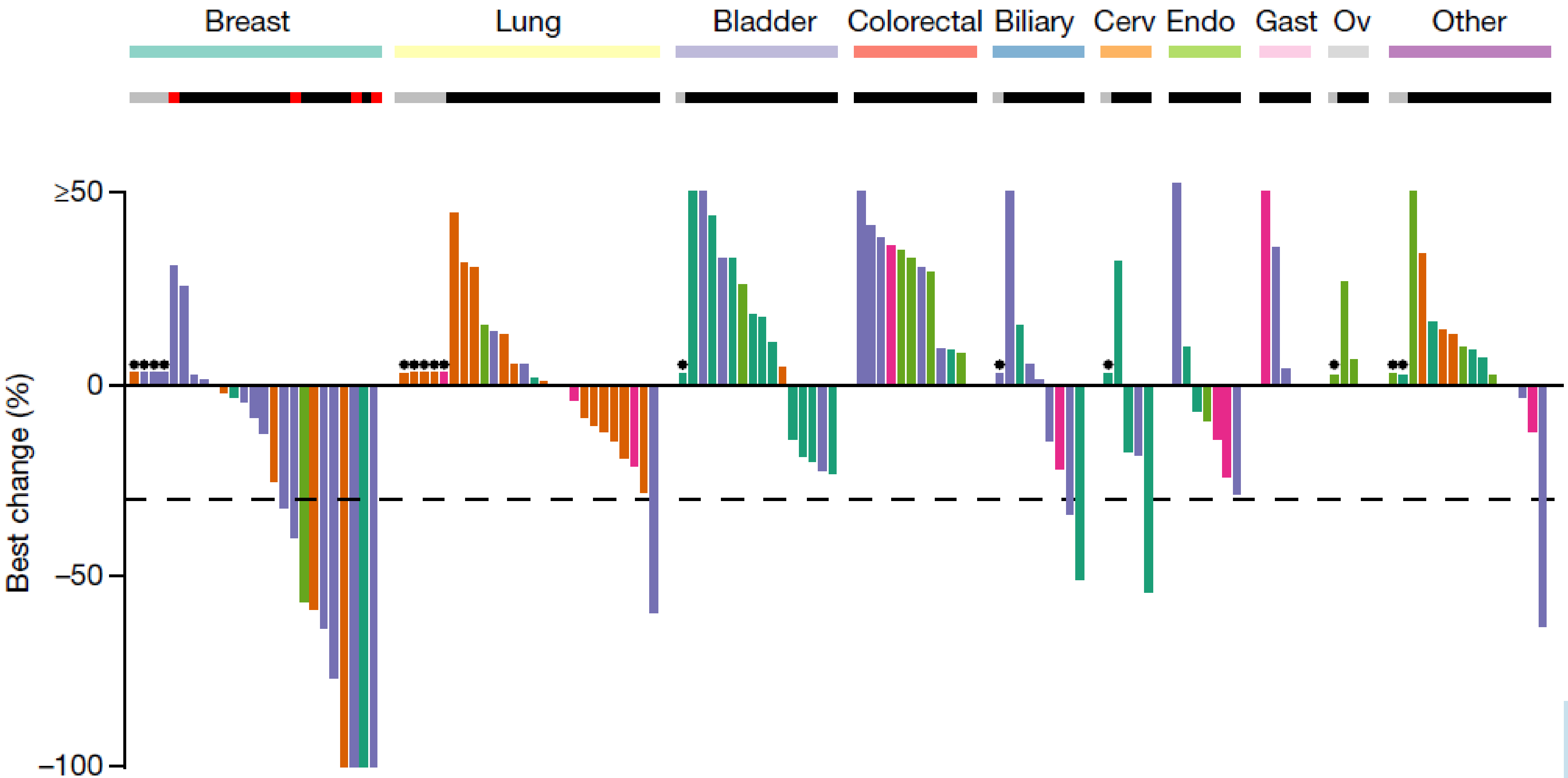
```
install.packages("basket")
```

Kane, M. J., Chen, N., Kaizer, A. M., Jiang, X., Xia, H. A., & Hobbs, B. P. (2020). Analyzing Basket Trials under Multisource Exchangeability Assumptions. *The R Journal*, 12(2).

The SUMMIT trial tested neratinib in HER2- and HER3-mutant tumors in 10 subtypes

- ORR ≤10% unacceptable, ORR ≥30% acceptable
- Independent Simon's optimal two-stage designs
  - Enroll 7 patients
  - If at least 1 response, enroll additional 11
  - If 4 responses in 18 total, reject null

Hyman, D. M. et al. (2018). HER kinase inhibition in patients with HER2- and HER3-mutant cancers. *Nature*, *554*(7691), 189–194.



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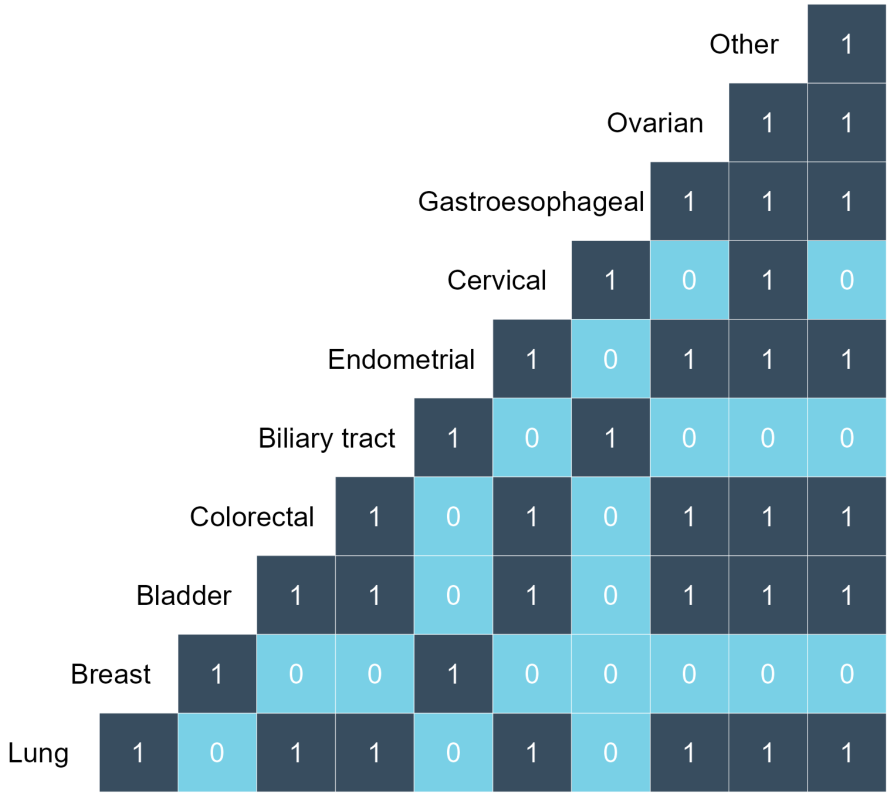
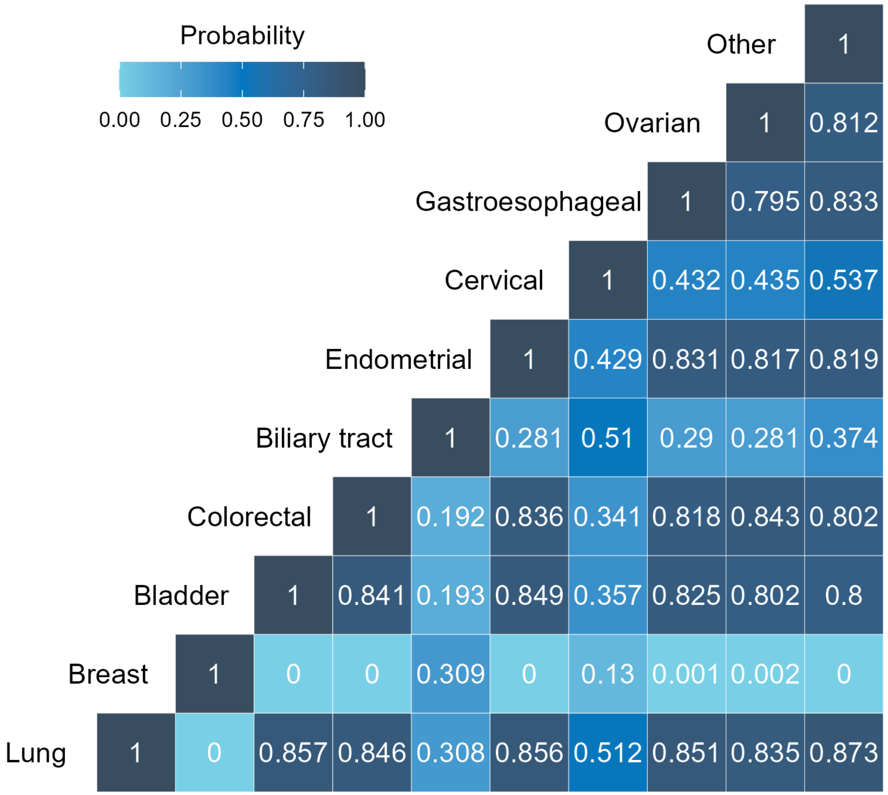
Maximum A Posteriori MEM



Posterior Exchangeability Probability



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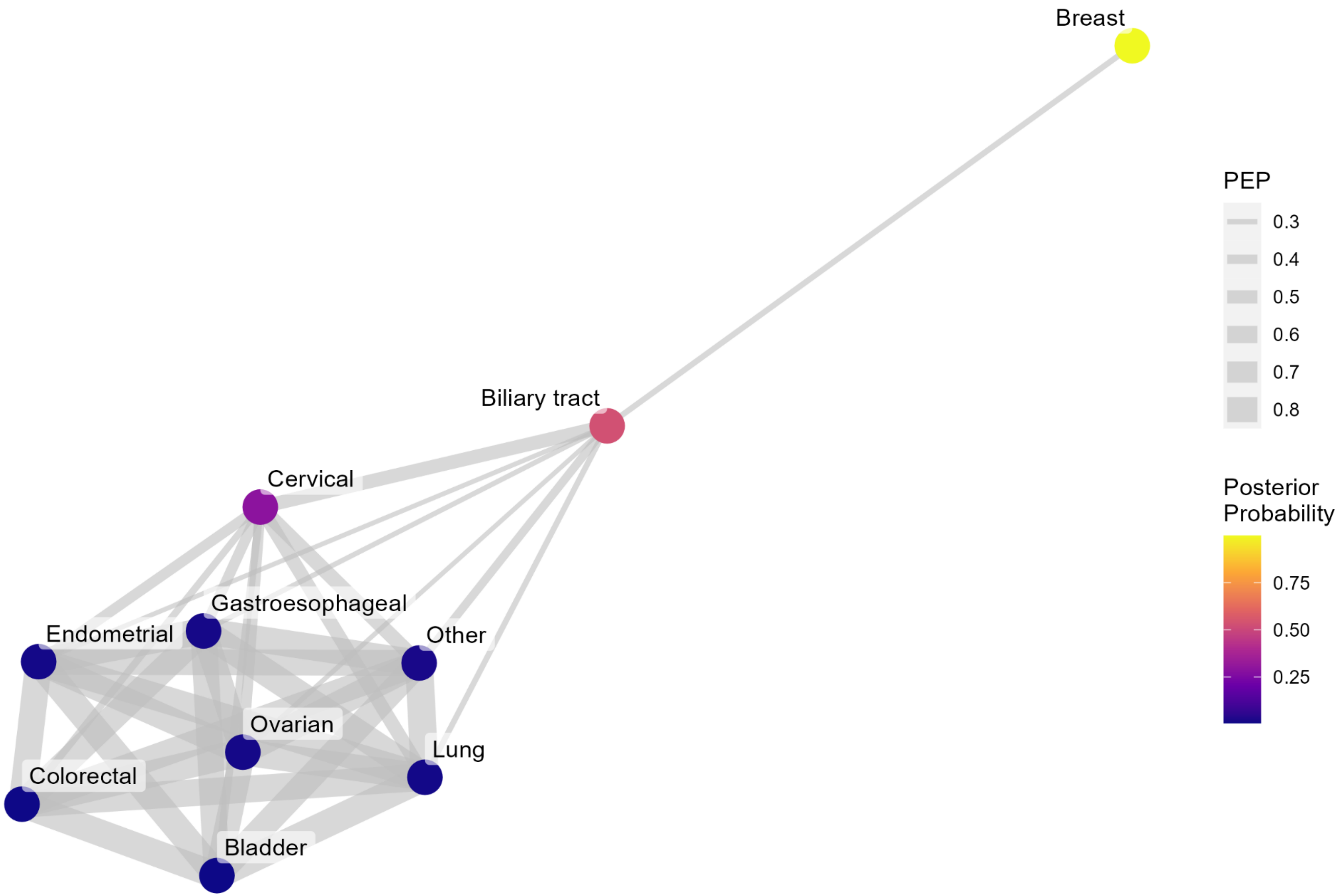

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False Discovery Rate (FDR) control:

1. Obtain the posterior probability that each basket j exceeds the null response rate, $\Pr(\pi_j > \pi_0|\mathbf{S})$
2. Order the posterior probabilities from largest to smallest, $\Pr(\pi_j > \pi_0|\mathbf{S})_{(1)}, \cdots, \Pr(\pi_j > \pi_0|\mathbf{S})_{(J)}$
3. For a given threshold of posterior probability, ϕ, identify the largest k such that $\Pr(\pi_j > \pi_0|\mathbf{S})_{(k)} > \frac{k}{J} \times \phi$
4. Declare all baskets $\Pr(\pi_j > \pi_0|\mathbf{S})_{(i)}, i = 1, \cdots, k$ to be significant at threshold ϕ

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Basket	N	ESS	PP	Rank (k)	$\frac{k}{J} \times \phi$	Significant*
Breast	25	27.9	0.999	1	0.095	TRUE
Biliary tract	11	16.0	0.531	2	0.190	TRUE
Cervical	5	15.4	0.291	3	0.285	TRUE
Other	19	91.7	0.020	4	0.380	FALSE
Gastroesophageal	7	90.4	0.016	5	0.475	FALSE
Ovarian	5	90.8	0.015	6	0.570	FALSE
Endometrial	8	94.6	0.013	7	0.665	FALSE
Lung	26	97.5	0.012	8	0.760	FALSE
Colorectal	17	100.0	0.007	9	0.855	FALSE
Bladder	18	102.0	0.007	10	0.950	FALSE

*At posterior threshold $\phi = 0.95$

Table 1. True response probabilities used to compare trial operating characteristics between Bayesian design with MEM and independent frequentist design.

	Basket									
Scenario	Lung n=26	Breast n=25	Bladder n=18	CRC n=17	Biliary n=11	EC n=8	Cervical n=5	GE n=7	Ovarian n=5	Other n=19
Global Alt	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30
Mixed Alt 1	0.10	0.30	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10
Mixed Alt 2	0.10	0.30	0.10	0.10	0.30	0.10	0.10	0.10	0.10	0.10
Mixed Alt 3	0.10	0.30	0.10	0.10	0.30	0.10	0.30	0.10	0.10	0.10
Global Null	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10

Alt = Alternative; CRC=Colorectal; EC=Endometrial; GE=Gastroesophageal

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Table 2. Power, false positive rate (FPR) and family-wise error rate (FWER) for each scenario using the MEM approach.

	FDR = 0.1			FDR = 0.05			FWER = 0.05	
Setting	Power	FPR	FWER	Power	FPR	FWER	Power	FPR
Global Null	–	0.10	0.32	–	0.04	0.17	–	0.01
Mixed Alternative 1	0.86	0.18	0.52	0.76	0.09	0.31	0.56	0.02
Mixed Alternative 2	0.81	0.24	0.58	0.69	0.12	0.38	0.48	0.03
Mixed Alternative 3	0.75	0.27	0.58	0.62	0.15	0.41	0.41	0.04
Global Alternative	0.96	–	–	0.94	–	–	0.90	–

*Posterior threshold 0.867 controls the FDR at 0.1; posterior threshold 0.946 controls the FDR at 0.05

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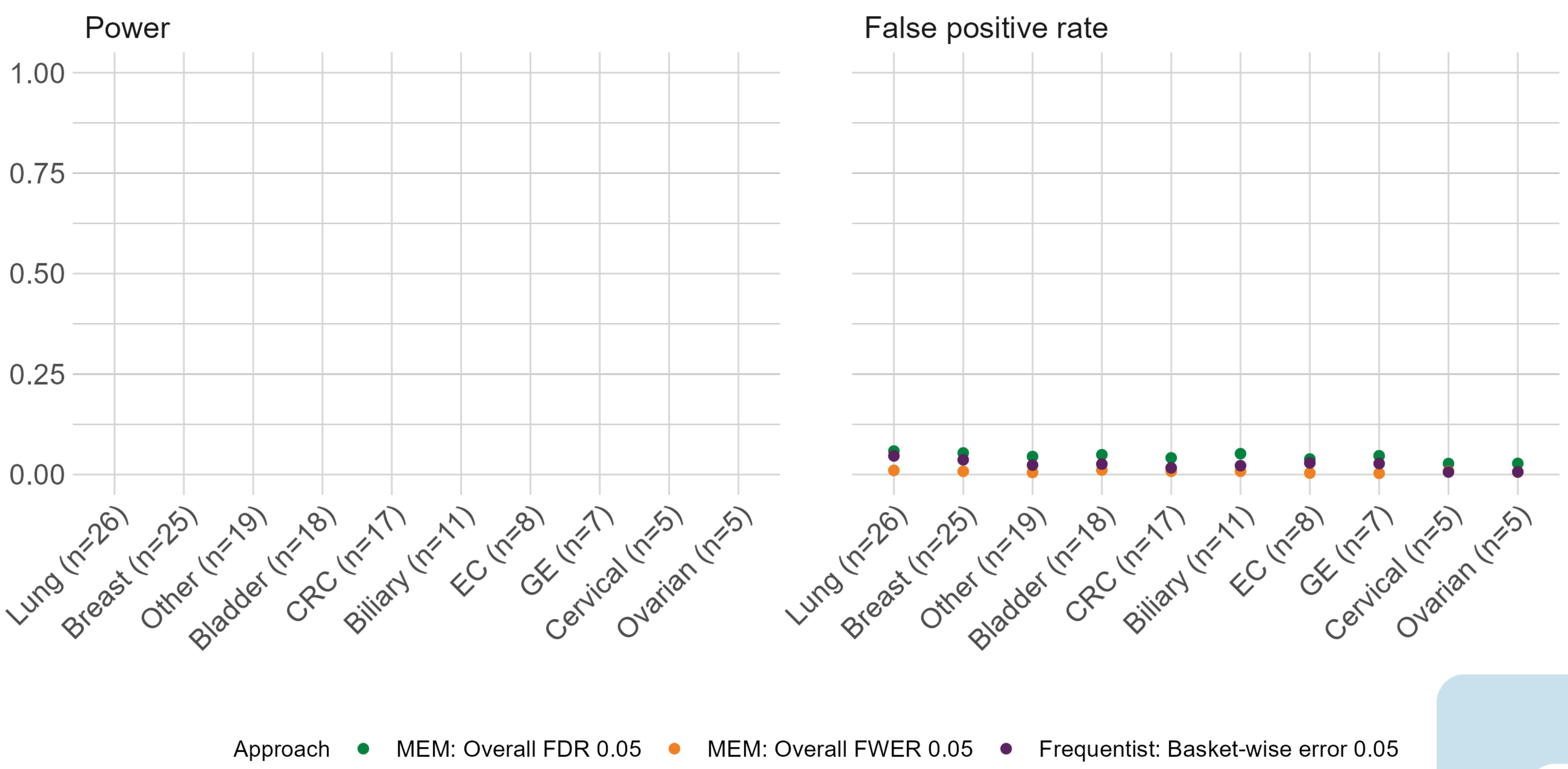
Global Null



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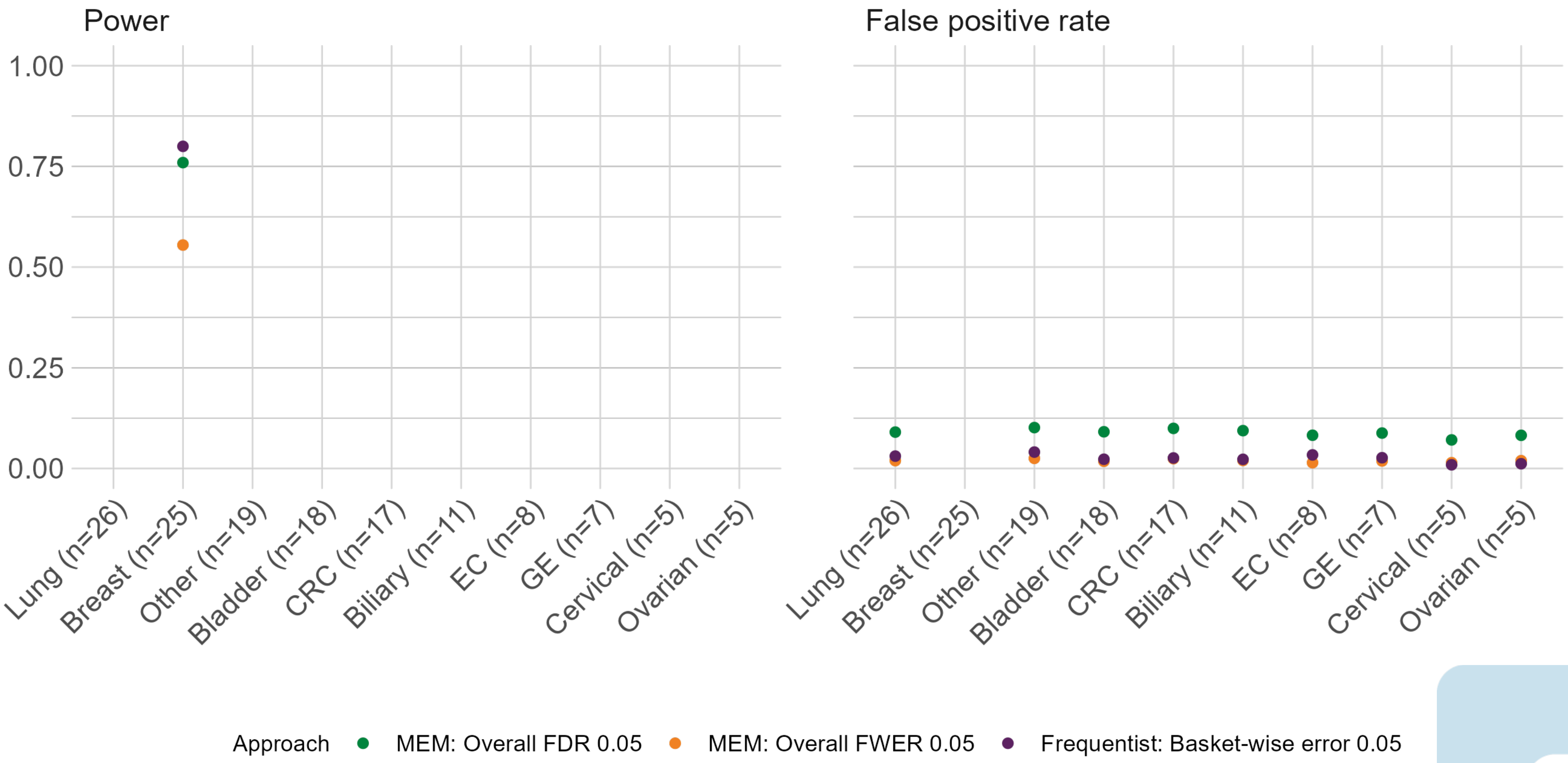
Mixed Alternative 1



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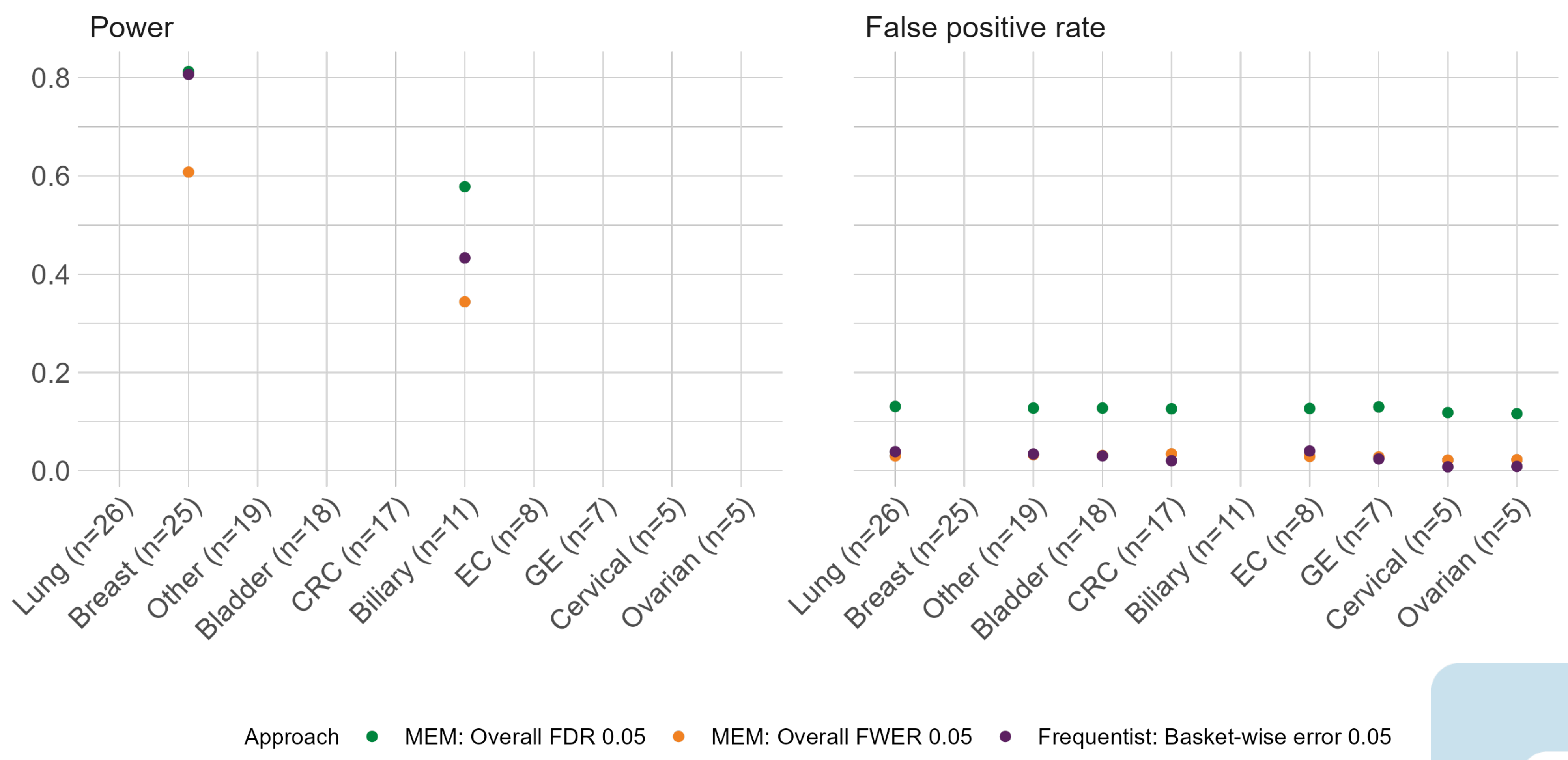
Mixed Alternative 2



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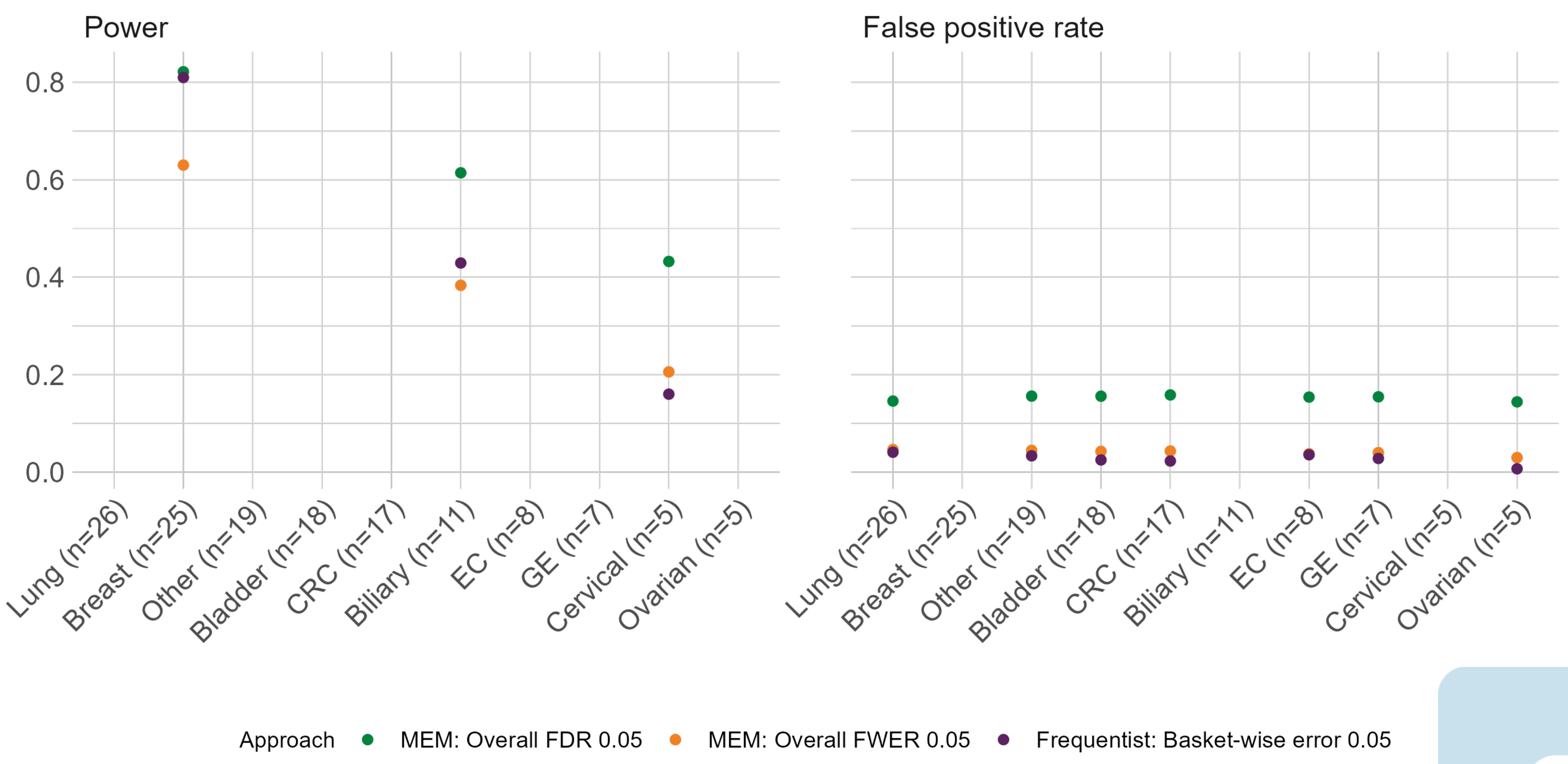
Mixed Alternative 3



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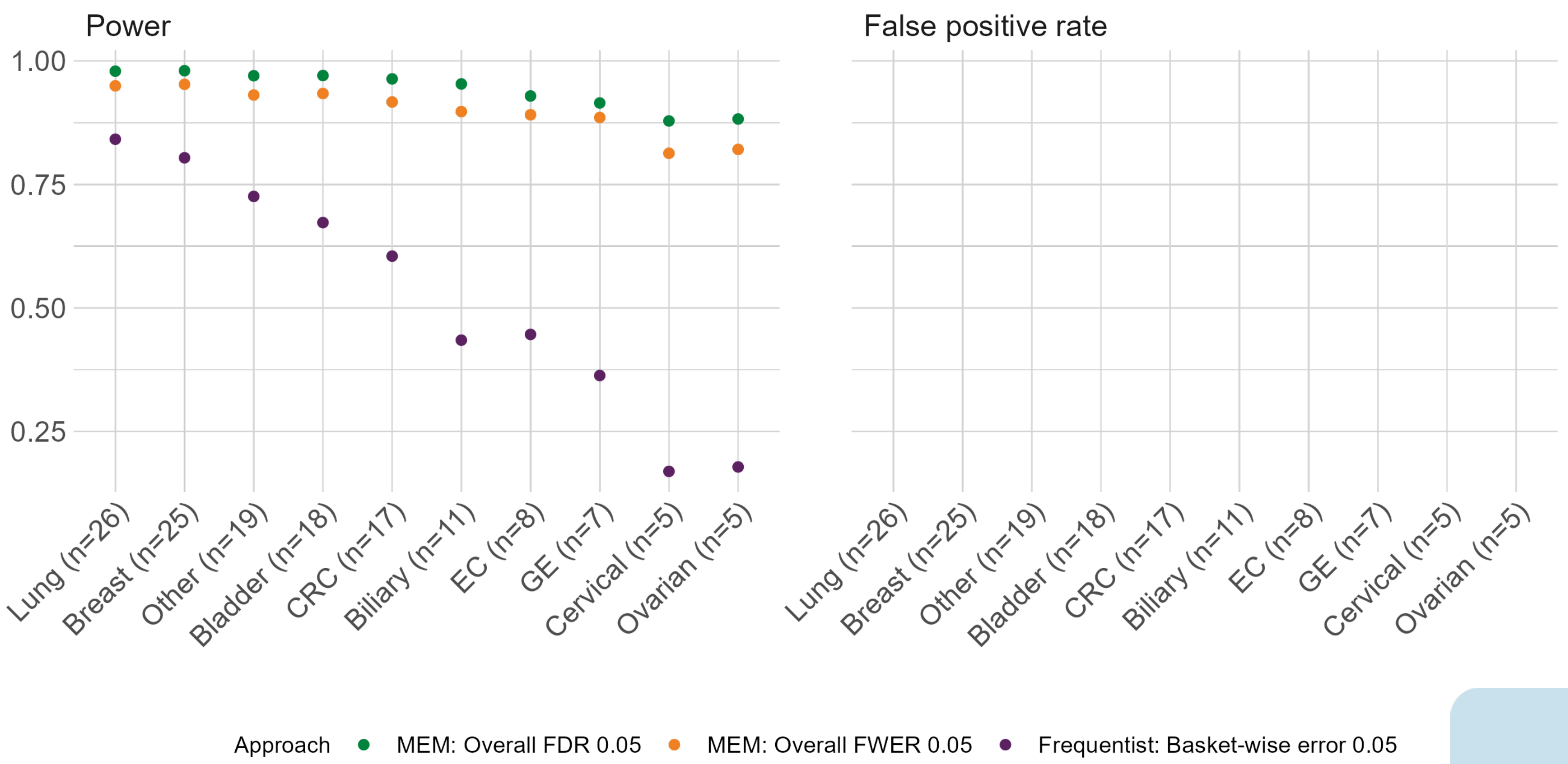
Global Alternative



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Summary

- MEM information-sharing compensates for small basket sample sizes and improves power
- MEM directly examines heterogeneity across baskets through posterior exchangeability probabilities
- FDR control maintains desired higher power at the cost of higher false positive rate
- Easy implementation with the basket package in R

Questions?

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@zabormetrics

@zabore

References

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